Weekly Quiz (Lec. 1: amino acids & proteins)

1. Which of the following is the simplest amino acid with a hydrogen atom as its side chain?

A. Alanine

B. Glycine

C. Valine

D. Serine

2. The peptide bond is formed between which two functional groups of amino acids?

A. Amino group and R group

B. Carboxyl group and R group

C. Amino group and carboxyl group

D. R group and R group

3. A polypeptide chain consisting of 45 amino acids is classified as:

A. Dipeptide

B. Oligopeptide

C. Polypeptide

D. Protein

4. Which level of protein structure is preserved during denaturation?

A. Primary structure

B. Secondary structure

C. Tertiary structure

D. Quaternary structure

5. Which of the following is an essential amino acid?

A. Alanine

B. Glutamate

C. Leucine

D. Aspartate

6. Which amino acid is purely ketogenic?

A. Leucine

B. Isoleucine

C. Phenylalanine

D. Tyrosine

7. The most abundant protein in the human body is:

A. Albumin

B. Globulin

C. Collagen

D. Hemoglobin

8. Which of the following amino acids is classified as both ketogenic and glucogenic?

A. Leucine

B. Lysine

C. Valine

D. Alanine

9. Which of the following is the correct order of protein structure organization from simplest to most complex?

A. Primary → Tertiary → Secondary → Quaternary

B. Secondary → Primary → Tertiary → Quaternary

C. Primary → Secondary → Tertiary → Quaternary

D. Tertiary → Secondary → Primary → Quaternary

10. The ubiquitin-proteasomal pathway primarily degrades proteins in the:

A. Lysosome

B. Cytosol

C. Mitochondria

D. Nucleus

11. In collagen, every third amino acid in the primary sequence is:

A. Proline

B. Hydroxyproline

C. Glycine

D. Alanine

12. Which of the following proteins is responsible for maintaining oncotic pressure?

A. Globulin

B. Fibrinogen

C. Albumin

D. Hemoglobin

13. A 65-year-old male presents with progressive memory loss and cognitive decline. Brain biopsy reveals extracellular accumulation of β-pleated fibrils. Which of the following proteins is most likely involved in this patient's condition?

A. α-synuclein

B. Amyloid β-protein (Aβ)

C. Prion protein

D. Huntingtin protein

14. A 3-year-old child presents with frequent bone fractures, blue sclerae, and hearing loss. Genetic testing reveals a mutation in the *COL1A1* gene. Which of the following is the most likely diagnosis?

A. Scurvy

B. Ehlers-Danlos syndrome

C. Osteogenesis imperfecta

D. Marfan syndrome

15. A 45-year-old sailor on a long sea voyage without access to fresh fruits and vegetables presents with bleeding gums, easy bruising, and poor wound healing. Which of the following is the underlying biochemical defect?

A. Defective collagen cross-linking due to copper deficiency

B. Impaired hydroxylation of proline and lysine residues in collagen

C. Mutations in *COL1A1* gene

D. Deficiency of lysyl oxidase

16. A 20-year-old female presents with joint hypermobility, skin hyperextensibility, and abnormal scarring. Genetic analysis reveals a mutation in the *COL5A1* gene. Which of the following is the most likely diagnosis?

A. Osteogenesis imperfecta

B. Scurvy

C. Ehlers-Danlos syndrome

D. Marfan syndrome

17. A 55-year-old male with a history of chronic alcoholism presents with jaundice, ascites, and confusion. Liver biopsy shows accumulation of misfolded proteins within hepatocytes. Which of the following mechanisms is most likely impaired in this patient?

A. Substrate-level phosphorylation

B. Ubiquitin-proteasomal degradation

C. Oxidative phosphorylation

D. Protein synthesis in the rough endoplasmic reticulum

18. A patient with α1-antitrypsin deficiency presents with cirrhosis and emphysema. This condition is best classified as which of the following?

A. A defect in collagen synthesis

B. A disorder of protein misfolding

C. A defect in protein glycosylation

D. A disorder of amino acid transport

Answers

1. B	7. C	13. B
2. C	8. B	14. C
3. C	9. C	15. B
4. A	10. B	16. C
5. C	11. C	17. B
6. A	12. C	18. B